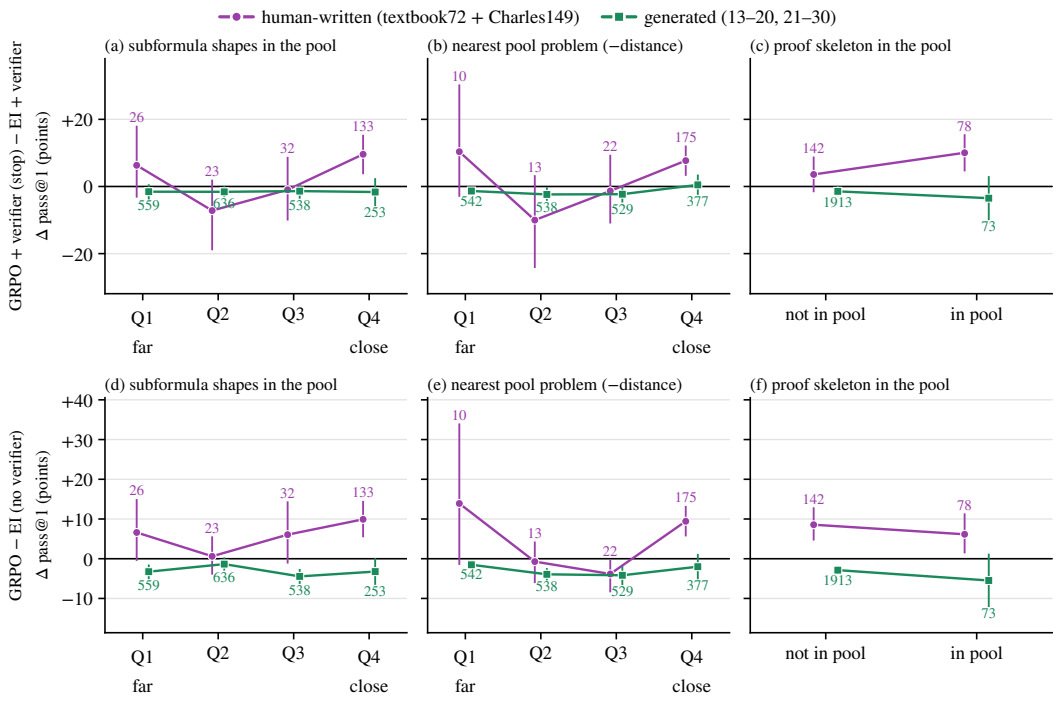
human-written (textbook72 + Charles149)
generated (13–20, 21–30)
(a) subformula shapes in the pool
(b) nearest pool problem (−distance)
(c) proof skeleton in the pool
GRPO + verifier (stop) − EI + verifier
Δ pass@1 (points)
+20
0
−20
26
559
23
636
32
538
133
253
10
542
13
538
22
529
175
377
142
1913
78
73
Q1
far
Q2
Q3
Q4
close
not in pool
in pool
(d) subformula shapes in the pool
(e) nearest pool problem (−distance)
(f) proof skeleton in the pool
GRPO − EI (no verifier)
Δ pass@1 (points)
+40
+30
+20
+10
0
−10
26
559
23
636
32
538
133
253
10
542
13
538
22
529
175
377
142
1913
78
73
Q1
far
Q2
Q3
Q4
close
not in pool
in pool
closeness to the RL curriculum pools (L2, L3; quartiles over all four sets); numbers: problems per point; bars: 95% intervals